SHAMAR ALLEN

Senior Software Engineer

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SKILLS

PROGRAMMING LANGUAGES

JavaScript, TypeScript, Python, SQL

BACKEND TECHNOLOGIES

Node.js, Express.js, NestJS, TypeORM, Sequelize

FRONTEND TECHNOLOGIES

React, Next.js, Redux, Recoil, TanStack Query, RxJS, HTML5, CSS3, BootStrap, TailwindCSS, MUI

DATABASE MANAGEMENT

PostgreSQL, MySQL, MongoDB, Redis

API DEVELOPMENT & INTEGRATION

RESTful APIs, GraphQL, tRPC, JSON-RPC, RabbitMQ, Kafka, Stripe, PayPal

CLOUD TECHNOLOGIES

AWS, Lambda, API Gateway, S3, RDS, Aurora, EC2, ECS, EKS, GCP, Cloud SQL, Cloud Storage, Cloud Run, Apigee, Cloud CDN

DEVOPS & CI/CD

Git, GitHub Actions, Jenkins, Docker, Kubernetes, Terraform

LLM TECHNOLOGIES

RAG, VectorDB, Semantic Search, Graph based GenAI, LangChain, LangGraph, MCP

SUMMARY

Senior Full Stack Engineer with 10+ years of experience building highly structured monolithic backend systems using TypeScript, Node.js, and NestJS, and responsive frontend applications with React and Next.js. Skilled in designing B2B SaaS platforms, multi-tenant systems, microservices, and cloud-native applications. Extensive experience in the healthcare domain, including FHIR-compliant data storage and HIPAA-compliant engineering best practices.

Proven A-Z product builder, capable of translating ambiguous, non-technical requirements into robust solutions.

Accomplished unofficial team lead, excelling in Agile, mentoring peers, fostering collaboration, and consistently earning top recognition while delivering high-quality, scalable, and secure systems.

PROFESSIONAL EXPERIENCE

SENIOR SOFTWARE ENGINEER — MAR 2022 – SEP 2025

Bask Health

Built a highly efficient end-to-end telehealth SaaS platform aiming to establish a Shopify-like infrastructure in the healthcare domain, reaching 500,000 active patients within the first year and generating $10M in revenue through hybrid per-clinic and per-patient pricing models.

- Led the end-to-end development of a high-performance telehealth SaaS platform using NestJS, Next.js, and AWS cloud infrastructure, working directly with the CEO and CTO to translate ambiguous strategic requirements into actionable engineering solutions that shaped key product and technical decisions.
- Architected a HIPAA-compliant monolithic backend using TypeScript and NestJS with a multi-tenant, event-driven design, leveraging modular architecture, dependency injection, and centralized error handling for secure and maintainable code.
- Designed highly efficient data storage on AWS Aurora PostgreSQL using multi-tenant schema design, and leveraged S3 strategies to manage various JSON serializations for the flow builder and visual customizer.
- Implemented a seamless integration between SQL database and AWS HealthLake for patient data, supporting both HL7 and FHIR-compliant interfaces to enable easy clinic integrations with other hospitals.
- Designed and documented well-structured public APIs, including RESTful and GraphQL endpoints, allowing clinics to manage assets and perform custom activities efficiently.
- Built a responsive frontend application using Next.js 15 with SSR, SSG, app routing, and TailwindCSS; developed a structured questionnaire visual customizer and a flow builder using React Flow, managing global state with Redux.
- Provisioned cost-efficient cloud infrastructure using AWS ECS, API Gateway, CloudFront, and CDN technologies for frontend deployment.
- Maintained a clean GitHub codebase by enforcing GitFlow and implementing secure CI/CD pipelines with GitHub Actions, including automated testing and encrypted secrets management for HIPAA-compliant deployments.
- Integrated payment gateways such as Stripe and PayPal, supporting both direct payments and connected accounts for clinics.
- Followed best QA practices, implementing comprehensive unit and end-to-end tests with Jest.
- Coordinated tasks and managed technical documentation on Jira and Confluence while unofficially leading platform development in a Kanban-based Agile environment.
- Collaborated directly with the Customer Support team to address real-time requests promptly, maintaining a 4.9-star support rating.

QUALITY ASSURANCE & TESTING

Jest, Cypress, Selenium, Mocha

MONITORING, LOGGING & DEBUGGING

Prometheus, Grafana, DataDog, Sentry.io

SEARCH & INDEXING

ElasticSearch, OpenSearch

METHODOLOGIES & SOFT SKILLS

Agile, Scrum, Kanban, SDLC, Collaboration, Mentorship, Leadership

TOOLS & TECHNOLOGIES

Visual Studio Code, Cursor, ChatGPT, ESLint, Prettier

EDUCATION

MASTER OF SCIENCE IN COMPUTER SCIENCE

Kean University
Sep 2017 – July 2019

BACHELOR OF ARTS IN COMPUTER SCIENCE

Anderson University
Aug 2010 – May 2014

CERTIFICATES

MICROSOFT CERTIFIED: AZURE DEVELOPER ASSOCIATE (AZ-204)

Microsoft | 2023

NODE.JS CERTIFICATION

OpenJS Foundation | 2024

SOFTWARE ENGINEERING CAREER TRACK

DataCamp | 2024

FULL STACK ENGINEER — SEP 2019 – FEB 2022

Pave Finance

- Contributed to the initial MVP development of Pave Pro, an AI-driven Investment SaaS, working in a fast-paced startup setting to drive product launch and continue scaling post-release.
- Engineered a scalable backend ecosystem with TypeScript, Express.js, and TypeORM, applying Hexagonal Architecture principles for clean separation of concerns.
- Architected optimized data persistence solutions leveraging Google Cloud SQL (MySQL) and Google Cloud Storage to ensure high performance and reliability.
- Implemented multi-tenant RBAC models and secure payment workflows, strengthening platform-wide data protection and compliance.
- Delivered a responsive web application using React 16 and TailwindCSS, with Redux for state management and TanStack React Query for API handling.
- Designed and provisioned cost-effective cloud infrastructure on Google Cloud (Cloud Run, Apigee, Cloud Storage, Cloud CDN), enabling seamless deployment and scalability.
- Maintained a well-structured GitHub repository, ensuring readability, code consistency, and adherence to modern development standards.
- Ensured product quality and stability by implementing unit and end-to-end test coverage with Jest, following industry QA best practices.
- Streamlined collaboration and delivery through Agile workflows, managing tasks via Trello and maintaining technical documentation in Confluence.

FULL STACK DEVELOPER — NOV 2015 – SEP 2017

MedZed

- Played a key role in designing and optimizing a scalable Patient Management System, leveraging Node.js and React to ensure adaptability to evolving requirements and maintainability over time.
- Enhanced backend efficiency by implementing robust APIs using Node.js, Express.js, and Sequelize, enabling seamless data access for high-demand operations.
- Refined PostgreSQL database schema and query structures to reduce response times significantly, improving performance under heavy user loads.
- Built and deployed serverless functions with AWS Lambda, streamlining data processing workflows, and integrated AWS S3 for secure and scalable file storage solutions.
- Improved frontend performance by incorporating lazy loading techniques and optimizing state management in React, ensuring a responsive and user-friendly interface.
- Championed quality assurance by utilizing Jest to maintain comprehensive test coverage and implemented Agile methodologies with Atlassian Jira for efficient task tracking and issue resolution.

JUNIOR NODE.JS DEVELOPER — JUN 2014 – NOV 2015

Blue Collar Brain

- As a key contributor to the Education Project, implemented MVC architecture and object-oriented programming (OOP) principles in JavaScript, Node.js, and EJS, optimizing transaction speeds, reducing maintenance costs, and enabling seamless scalability for future growth.
- Developed and sustained a high-performance backend using JavaScript and Express.js, ensuring swift and reliable API performance to support critical application features.
- Enhanced the database layer by optimizing MySQL schema and queries, resulting in faster data retrieval and scalable transaction support, even during peak usage.
- Designed and deployed RESTful APIs with JavaScript and Express.js to ensure smooth data flow and facilitate seamless integrations with external systems, significantly improving response times and interoperability.